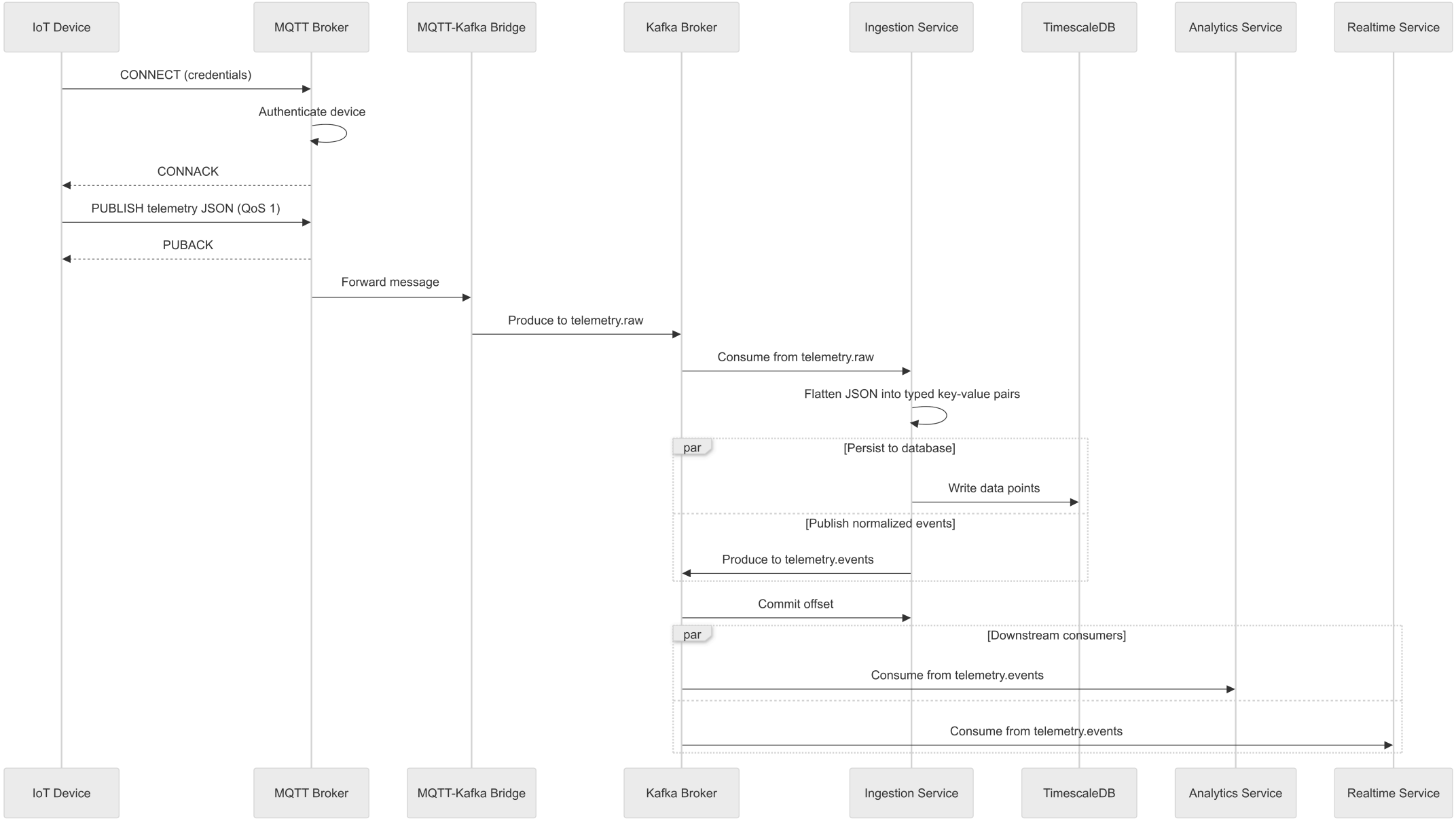
IoT Device
MQTT Broker
MQTT-Kafka Bridge
Kafka Broker
Ingestion Service
TimescaleDB
Analytics Service
Realtime Service
CONNECT (credentials)
Authenticate device
CONNACK
PUBLISH telemetry JSON (QoS 1)
PUBACK
Forward message
Produce to telemetry.raw
Consume from telemetry.raw
Flatten JSON into typed key-value pairs
par
[Persist to database]
Write data points
[Publish normalized events]
Produce to telemetry.events
Commit offset
par
[Downstream consumers]
Consume from telemetry.events
Consume from telemetry.events
IoT Device
MQTT Broker
MQTT-Kafka Bridge
Kafka Broker
Ingestion Service
TimescaleDB
Analytics Service
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